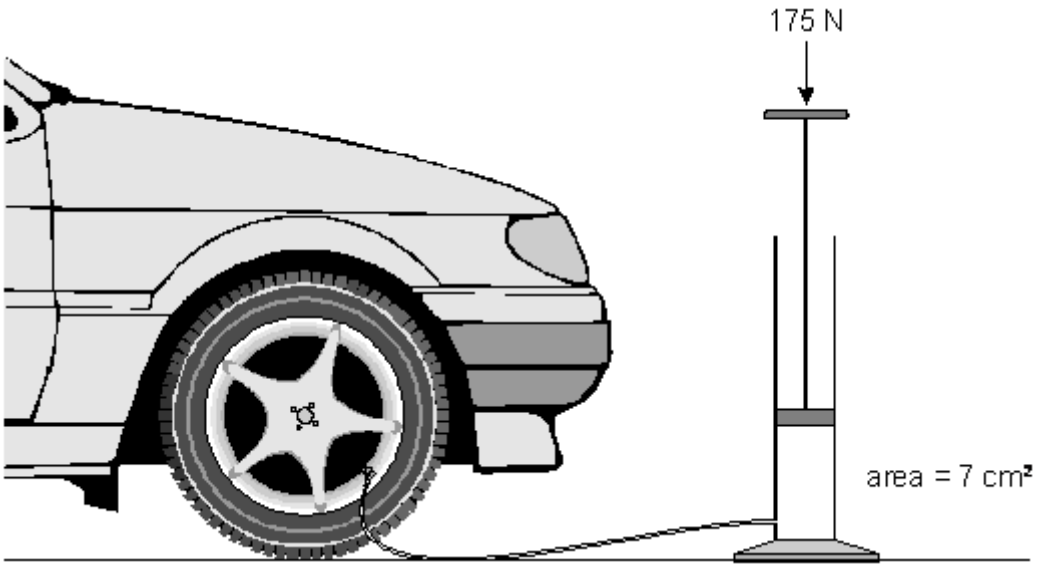
Q1. Karen wants to pump up her car tyre.
Her pump has a piston with an area of 7 cm^2.



Karen pushes the handle down with a force of 175 N.

(a) What pressure does she exert on the air in the pump?

.....

..... N/cm^2

1 mark

(b) The air pressure in the tyre is 27 N/cm^2.
What pressure would be needed **in the pump** in order to pump more air into the tyre?

.....

.....

1 mark

(c) Another of Karen's car tyres exerts a pressure of 30 N/cm^2 on the road. The area of the tyre in contact with the road is 95 cm^2
What is the force exerted by the tyre on the road?

..... N

1 mark

Maximum 3 marks

Q1.

(a) 25

accept '175 ÷ 7'

1 (L7)

(b) any **one** from

- greater than 27 N/cm^2

  the unit is required for the mark

  *do **not** accept '27 N/cm^2'*

- greater than the pressure in the tyre

  accept any answer greater than 27 N/cm^2

1 (L7)

(c) 2850

1 (L7)

[3]